The Constitutional Doctrine of Biosecurity Evidence Preservation

A Bayesian Actor-Network Model of Strategic Biological Uncertainty

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Abstract

This paper develops a mathematical and statistical framework for a constitutional doctrine of biosecurity evidence preservation. The doctrine is designed for strategic biological events in which early scientific evidence becomes missing, suppressed, altered, restricted, or selectively disclosed. The framework does not presume that any particular pandemic or outbreak is biological warfare. Instead, it formalizes how a state should preserve evidence, screen for financial and procurement foreknowledge, distinguish ordinary crisis mobilization from suspicious convergence, and reserve attribution for cases satisfying a strict evidentiary threshold. The model combines Bayesian inference, event-study finance, procurement anomaly detection, actor-network analysis, constitutional political theory, information warfare, and biosecurity governance. Its central claim is procedural: when early outbreak-relevant evidence becomes unavailable, the first duty of the state is not retaliation, accusation, or censorship, but evidence preservation under constitutional control.

The paper ends with "The End"

Contents

List of Figures

List of Tables

1 Introduction

Modern biological events are not merely epidemiological events. They can become public-health crises, financial shocks, geopolitical conflicts, information-warfare theaters, industrial-policy accelerators, and constitutional stress tests. A state facing a high-impact outbreak may not know whether it is observing natural spillover, laboratory accident, negligent biosafety failure, opportunistic profiteering, information warfare, or deliberate biological attack. The difficulty is not only that the truth is uncertain. The difficulty is that the evidence needed to resolve the truth may itself be perishable, suppressed, traded upon, classified, manipulated, or destroyed.

This paper proposes the *Constitutional Doctrine of Biosecurity Evidence Preservation*. The doctrine is activated by strategic biological uncertainty, not by a conclusion of biological warfare. Its minimum trigger is the material loss, suppression, alteration, restriction, or unexplained non-disclosure of early outbreak-relevant scientific data. Once triggered, the doctrine requires immediate preservation of biological, clinical, procurement, financial, logistical, communications, and vaccine-platform records, including foreign-linked contracts and collaborations. Parliament is the primary constitutional authority; the President is a temporary backstop if Parliament fails; the constitutional court reviews legality; and a later biosecurity tribunal evaluates technical attribution.

The paper has three aims. First, it defines an actor-network model for strategic biological attribution. Second, it develops a statistical pipeline for detecting pre-disclosure financial and procurement anomalies. Third, it proposes a constitutional decision rule that separates investigation, counter-narration, sanction, and state-level attribution.

The analysis is deliberately conditional. It does not assert that COVID-19 was biological warfare. Public assessments remain contested on origin and generally do not establish a biological-weapon conclusion [3, 4, 5]. The framework instead asks: if states must reason under strategic biological uncertainty, what evidentiary and constitutional structure prevents both naivete and reckless accusation?

2 Factual and Epistemic Status

Let

$$D_0 = \text{the first international disclosure date of a strategic biological event.}$$

For COVID-19, a defensible public anchor is 31 December 2019, when Chinese authorities reported a cluster of pneumonia cases in Wuhan and the WHO China Country Office was informed of pneumonia of unknown etiology [1, 2]. This date is not necessarily the date of first infection, first internal awareness, first laboratory observation, or first market recognition. It is a public procedural anchor.

The doctrine distinguishes four layers:

1. biological/material origin;
2. information warfare and attribution conflict;
3. crisis-industrial policy and war-economy mobilization;
4. financial profiteering or foreknowledge.

Evidence from one layer may update beliefs about another, but no layer is allowed to substitute for all others. In particular, post-disclosure vaccine mobilization may show crisis-industrial policy, but it does not by itself establish biological warfare. Before COVID-19, coronavirus vaccine candidates and platforms existed, but the first WHO emergency-use validation for a COVID-19 vaccine occurred on 31 December 2020 [6]. Therefore, a strong pre-D_0 biological-warfare inference would require pathogen-specific or protective countermeasure evidence before public disclosure, not merely the existence of general coronavirus research.

3 Strategic Biological Uncertainty

Definition 1 (Strategic biological event). *A biological event is strategic if its scale, ambiguity, adversarial potential, or institutional consequences require state-level intervention beyond ordinary public-health, agricultural, veterinary, environmental, or military-medical authority.*

Definition 2 (Strategic biological uncertainty). *Strategic biological uncertainty exists when a strategic biological event occurs and early outbreak-relevant evidence is unavailable, unreliable, missing, suppressed, altered, restricted, or selectively disclosed in a way that impairs origin reconstruction, public-health response, financial-risk assessment, or state defense.*

Definition 3 (Actor-network). *An actor-network $a \in \mathcal{A}$ is a set of linked institutions, individuals, firms, laboratories, funds, state organs, contractors, logistics nodes, intermediaries, or procurement channels. The actor-network is the unit of inference. A nation, ethnicity, profession, migrant group, or scientific field is not an admissible unit of attribution.*

Definition 4 (Biological-warfare hypothesis). *For actor-network a, define*

$H_1(a)$ = actor-network a participated in, enabled, protected, or knowingly obstructed accountability for a biological-warfare event.

The null is

H_0 = no deliberate biological attack; anomalies arise from natural emergence, accident, ordinary preparedness, speculation, crisis policy, or noise.

4 Constitutional Doctrine

Axiom 1 (Evidence preservation before attribution). *When strategic biological uncertainty arises, the state must first preserve evidence rather than accuse, retaliate, suppress speech, or assign collective blame.*

Axiom 2 (Trigger). *The minimum trigger is material suppression, disappearance, alteration, restriction, or unexplained non-disclosure of early outbreak-relevant scientific data.*

Axiom 3 (Institutional sequence). *The trigger automatically notifies the Chief Public-Health Officer and the National Security Council. The Chief Public-Health Officer reports to the President. Parliament has the primary duty to issue a preservation-and-disclosure order. If Parliament fails, the President may issue temporary preservation directives, subject to constitutional-court review.*

Axiom 4 (No collective blame)**.** *No inference may be drawn against a population, ethnicity, migrant group, scientific profession, or nation solely because a suspicious actor-network is located within or linked to that category.*

The preservation order covers:

1. biological samples, freezer logs, lab notebooks, sequencing data, and biosafety records;
2. clinical records, pneumonia clusters, unusual deaths, serology, and hospital alerts;
3. procurement contracts, reagents, vials, syringes, PPE, cold-chain equipment, and sequencing supplies;
4. vaccine-platform, immunogen, antibody, animal-model, and challenge-study records;
5. financial trades, options positions, beneficial ownership, sovereign-fund movements, and insider transactions;
6. transport, customs, personnel movement, sample-transfer, and logistics records;
7. communications among laboratories, ministries, regulators, firms, journals, and public-health agencies;
8. foreign-linked contracts, grants, funding flows, personnel exchanges, sample transfers, licensing arrangements, and research collaborations.

5 Evidence Streams

Let the set of evidence streams be

$$\mathcal{S} = \{F, P, B, L, I\},$$

where F is financial foreknowledge, P is procurement-geographic anomaly, B is biological/material evidence, L is logistics/operational linkage, and I is information-control evidence.

Evidence from stream $s \in \mathcal{S}$ for actor-network a is denoted $E_{s,a}$. Define an admissibility indicator

$$A_{s,a} \in \{0, 1\}.$$

Definition 5 (Admissibility)**.** *Evidence $E_{s,a}$ is admissible for attribution if and only if it is:*

1. *pre-D_0;*
2. *costly to fake;*
3. *actor-specific.*

Formally,

$$A_{s,a} = 1$$

if all three conditions hold, and $A_{s,a} = 0$ otherwise.

The admissibility rule prevents hindsight selection. Post-D_0 evidence may be relevant to crisis response, obstruction, or information warfare, but it is not automatically attribution-grade.

6 Bayesian Attribution Core

The posterior odds for actor-network a are

$$\log \frac{\mathbb{P}(H_1(a) \mid E)}{\mathbb{P}(H_0 \mid E)} = \log \frac{\mathbb{P}(H_1(a))}{\mathbb{P}(H_0)} + \sum_{s \in \mathcal{S}} A_{s,a} \log LR_{s,a},$$

where

$$LR_{s,a} = \frac{\mathbb{P}(E_{s,a} \mid H_1(a))}{\mathbb{P}(E_{s,a} \mid H_0)}.$$

A prior of 5% corresponds to prior odds:

$$\frac{0.05}{0.95} = \frac{1}{19}.$$

The prior makes the hypothesis analytically live but does not justify attribution. Evidence must shift the posterior.

Lemma 1 (Neutral evidence)**.** *If $LR_{s,a} = 1$, then evidence stream s does not change posterior odds for actor-network a.*

Proof. If $LR_{s,a} = 1$, then $\log LR_{s,a} = 0$. Hence the term $A_{s,a} \log LR_{s,a}$ is zero regardless of admissibility. The posterior odds are unchanged by that stream. □

Proposition 1 (Weakness of ordinary crisis mobilization)**.** *If an evidence item E is highly probable under both biological warfare and ordinary pandemic crisis, then E has weak attribution value.*

Proof. Attribution value is governed by

$$LR = \frac{\mathbb{P}(E \mid H_1)}{\mathbb{P}(E \mid H_0)}.$$

If $\mathbb{P}(E \mid H_1) \approx \mathbb{P}(E \mid H_0)$, then $LR \approx 1$. By the neutral-evidence lemma, such evidence weakly updates or fails to update posterior odds. □

7 Financial Foreknowledge Model

Let

$$W_F = [D_0 - 14, D_0)$$

be the financial foreknowledge window. It is short enough to limit generic long-run speculation and long enough to detect event-proximate preparation.

For instrument j and date t, let $V_{j,t}$ be options volume. Estimate expected log-volume:

$$\log V_{j,t} = \alpha_j + \beta_j' X_t + \gamma_j' C_{j,t} + \varepsilon_{j,t},$$

where X_t includes market volatility, index returns, liquidity, macro news, and risk-free-rate conditions; $C_{j,t}$ includes earnings events, calendar effects, index rebalancing, merger rumors, and year-end effects.

Define abnormal options volume:

$$Z^F_{j,t} = \frac{\log V_{j,t} - \widehat{\mu}_{j,t}}{\widehat{\sigma}_j}.$$

Define pandemic-sector exposure:

$$\omega_j = \begin{cases} +1, & \text{vaccines, diagnostics, PPE, remote work, cloud, logistics, safe assets,} \\ -1, & \text{airlines, cruises, hotels, tourism, energy, commercial real estate,} \\ 0, & \text{neutral sectors.} \end{cases}$$

Let $q_{a,j,t}$ be traced exposure of actor-network a to instrument j at time t. The financial foreknowledge score is:

$$F_a = \sum_{t \in W_F} \sum_j q_{a,j,t} \omega_j Z^F_{j,t}.$$

A high F_a screens actor-networks for possible foreknowledge. It does not by itself establish biological warfare.

Financial stage	Statistical object	Evidentiary function
Screen	abnormal options volume	identifies possible foreknowledge
Classify	pandemic-shaped sector rotation	distinguishes crisis pattern from noise
Incriminate	linked-party transaction tracing	connects trades to actor-networks
Converge	pairing with biological/material stream	supports attribution-grade inference

Table 1: Financial-forensics pipeline

8 Procurement-Geographic Model

Let $Y_{i,r,t}$ denote procurement in region i, item class r, and date t, where

$$r \in \{\text{diagnostic reagents}, \text{cold-chain/vials}, \text{sequencing supplies}\}.$$

Assume a negative-binomial model:

$$Y_{i,r,t} \sim \mathrm{NegBin}(\lambda_{i,r,t}, \phi_r).$$

Let

$$\log \lambda_{i,r,t} = \alpha_i + \delta_r + \tau_t + \theta_1 \log(\text{population}_i) + \theta_2 \text{hospital density}_i + \theta_3 \text{respiratory burden}_{i,t} + \theta_4 \text{baseline budget}_{i,t}.$$

Define the procurement anomaly:

$$Z^P_{i,r,t} = \frac{Y_{i,r,t} - \widehat{\lambda}_{i,r,t}}{\sqrt{\widehat{\lambda}_{i,r,t}}}.$$

The regional procurement score is:

$$P_i = \sum_r \sum_{t \in W_F} w_r Z^P_{i,r,t}.$$

Map regions to actor-networks through a linkage matrix $G_{a,i}$:

$$P_a = \sum_i G_{a,i} P_i.$$

This model identifies candidate foreknowledge zones. It does not prove foreknowledge without linked-party tracing and biological confirmation.

9 Biological and Material Evidence

Biological/material evidence is the necessary attribution stream. Let

$$B_a = \sum_m \eta_m b_{a,m},$$

where $b_{a,m}$ indicates biological/material item m linked to actor-network a, and η_m is its evidentiary weight.

Evidence item	Timing	Attribution value
generic coronavirus research	pre-D_0	low
SARS-like sample records	pre-D_0	moderate to high
challenge studies linked to actor-network	pre-D_0	high
SARS-CoV-2-protective countermeasure	pre-D_0	very high
insider neutralizing serology	pre-D_0	very high
post-D_0 vaccine contracts	post-D_0	low for origin attribution

Table 2: Biological/material evidentiary hierarchy

10 Logistics and Operational Linkage

Let $R_{a,i}$ measure the operational linkage between actor-network a and region i:

$$R_{a,i} = f(\text{personnel travel}, \text{sample transfers}, \text{cargo routes}, \text{cold-chain movement}, \text{facility access}).$$

Let C_i be early infection-cluster intensity in region i. The logistics score is:

$$L_a = \sum_i R_{a,i} C_i.$$

A high L_a indicates overlap between an actor-network's logistical footprint and early biological events. This stream becomes powerful only when paired with B_a, F_a, or P_a.

11 Information-Control Evidence

Let M_t be a missing-evidence indicator:

$$M_t = \begin{cases} 1, & \text{early outbreak-relevant data are missing, suppressed, altered, restricted, or selectively disclosed,} \\ 0, & \text{otherwise.} \end{cases}$$

The constitutional trigger activates when:

$$M_t = 1 \quad \text{and} \quad \text{the missing data are material to origin reconstruction or public-health response.}$$

Information-control evidence may support a cover-up or strategic ambiguity hypothesis, but it cannot substitute for biological/material evidence.

12 Convergence Score

Define:

$$C_a = \kappa_F F_a + \kappa_P P_a + \kappa_B B_a + \kappa_L L_a + \kappa_I I_a.$$

The convergence score ranks actor-networks for investigation. Attribution requires a separate threshold.

Let

$$N_a = \sum_{s \in \mathcal{S}} A_{s,a} \mathbf{1}\{E_{s,a} \text{ is materially positive}\}.$$

Definition 6 (Attribution threshold)**.** *Biological-warfare attribution is permitted only if:*

$$N_a \geq 2,$$

and

$$A_{B,a} = 1, \qquad B_a > b^*.$$

Thus, at least two independent pre-D_0, costly-to-fake, actor-specific streams are required, including one biological/material stream.

Theorem 1 (No single-stream attribution)**.** *Under the attribution threshold, no single evidence stream can justify biological-warfare attribution.*

Proof. The threshold requires $N_a \geq 2$. Since N_a counts materially positive admissible streams, one stream can produce at most $N_a = 1$. Therefore no single stream satisfies the attribution threshold. Additionally, the biological/material stream must be positive and admissible, so financial, procurement, logistics, or information-control evidence alone cannot attribute biological warfare. □

13 Adversarial Bayesianism and Deception

Strategic adversaries may conceal evidence, fabricate evidence, plant decoys, or selectively disclose moderate convergence. Therefore the model includes a deception filter.

Definition 7 (Anti-decoy filter)**.** *The alleged attacker is more likely real than decoy if convergence evidence is:*

1. *pre-D_0;*
2. *costly to fake;*
3. *actor-specific.*

Moderate convergence is sufficient to trigger investigation and counter-narration. It is not sufficient for final attribution.

Observed pattern	Possible interpretation	Required response
moderate convergence around real network	true signal	investigate
moderate convergence around decoy network	false-flag signal	apply anti-decoy filter
unrelated anomalies	crisis noise	run placebo tests
selective disclosure	information-warfare artifact	preserve records and audit source

Table 3: Adversarial interpretation of convergence evidence

14 Decision Ladder

Let

$$\pi_a = \mathbb{P}(H_1(a) \mid E).$$

The posterior probability supports different constitutional actions.

Posterior range	Action
$5\% - 10\%$	monitor and include in threat model
$10\% - 25\%$	investigate and preserve evidence
$25\% - 50\%$	counter-narrate and demand coercive transparency
$50\% - 75\%$	tribunal inquiry and targeted restrictions
$75\% - 95\%$	targeted sanctions against named actors and institutions
$> 95\%$ plus attribution threshold	possible state-level action if state command/protection is shown

Table 4: Posterior decision ladder

Whole-state punishment requires evidence of state command, state protection, knowing enablement, or knowing obstruction beyond reasonable strategic doubt. Territorial suspicion is insufficient.

15 State Protection and Remedies

State protection is not mere secrecy. A state may legitimately classify or withhold sensitive data if it preserves records under sealed custody and provides a classified index for cleared review. State protection becomes suspicious when state organs block audits, threaten witnesses, manipulate courts, reward implicated networks, or destroy records after a preservation trigger.

Definition 8 (State protection). *State protection exists when state organs knowingly use legal, diplomatic, coercive, financial, or institutional power to prevent investigators from accessing attribution-relevant evidence or to preserve the position of implicated actors.*

Attribution level	Remedy	Limitation
named individuals	prosecution, asset freezes, travel bans	due process required
firms/labs/institutions	sanctions, debarment, license revocation	actor-specific evidence required
state agencies/factions	targeted sanctions, inspection demands	command/protection evidence required
whole state	treaty claims, reparations, state sanctions	beyond reasonable strategic doubt

Table 5: Remedy ladder

16 AI/ML Implementation

The empirical pipeline can be implemented using supervised, unsupervised, and causal-inference tools. The goal is not automatic attribution but candidate ranking.

Task	Model family	Output
options anomaly detection	event-study regression, isolation forest	abnormal volume score
sector rotation classification	penalized logistic regression, random forest	pandemic-shaped portfolio score
procurement anomaly detection	hierarchical count model, Bayesian shrinkage	regional procurement score
linked-party tracing	graph learning, community detection	actor-network map
textual signal extraction	sentiment analysis, topic model	information-control indicators
deception filtering	adversarial validation	decoy-risk score
posterior aggregation	Bayesian model averaging	actor-network posterior

Table 6: AI/ML components

17 Algorithm

Algorithm 1: Constitutional Biosecurity Evidence Preservation and Attribution Screening
1. Observe strategic biological event and set public anchor D_0.
2. Test whether early scientific data are missing, suppressed, altered, restricted, or selectively disclosed.
3. If material missing evidence exists, trigger notification to Chief Public-Health Officer and National Security Council.
4. Chief Public-Health Officer submits public report and classified annex to President.
5. Parliament issues preservation-and-disclosure order; if Parliament fails, President issues temporary preservation directives subject to constitutional-court review.
6. Preserve biological, clinical, procurement, financial, logistics, communications, and foreign-linked records.
7. Screen financial data in $W_F = [D_0 - 14, D_0)$ for abnormal options volume.
8. Classify sector rotation as pandemic-shaped or generic.
9. Trace beneficial ownership, counterparties, intermediaries, sovereign funds, private equity, and linked parties.
10. Screen procurement geography for abnormal diagnostic reagents, cold-chain/vials, and sequencing supplies.
11. Map procurement regions to actor-networks through contracts, suppliers, labs, hospitals, and logistics nodes.
12. Search for biological/material confirmation: samples, serology, countermeasures, challenge studies, lab records.
13. Apply admissibility filter: pre-D_0, costly to fake, actor-specific.
14. Compute convergence score C_a and posterior π_a.
15. Attribute biological warfare only if at least two independent admissible streams exist, including one biological/material stream.
16. Apply remedies: named actors and institutions first; whole-state remedies only with state command/protection beyond reasonable strategic doubt.

Table 7: Procedural algorithm

18 Vector Graphic

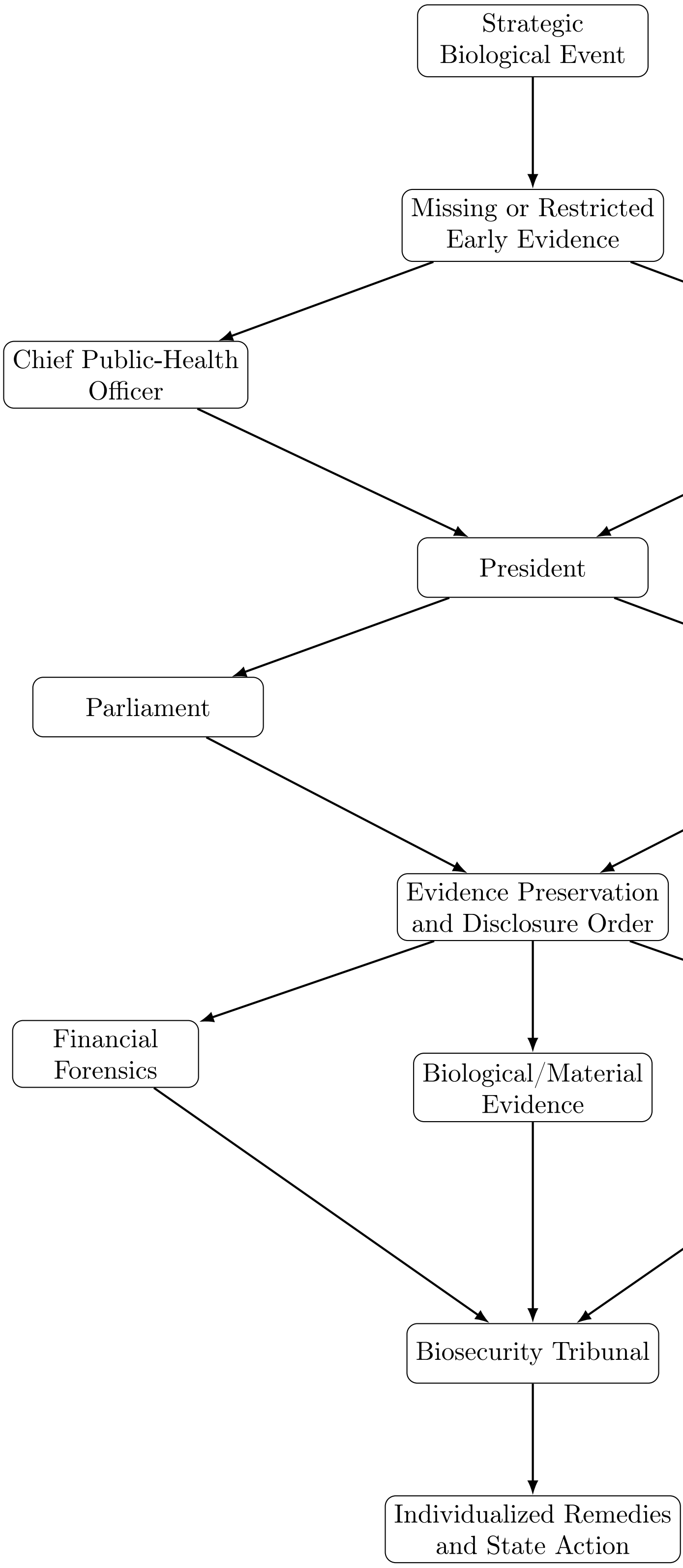

Figure 1: Constitutional and empirical architecture of the doctrine

19 Political Economy and Warfare Economics

Biological events can create asymmetric information in public-health, financial, procurement, and defense systems. If actors possess early knowledge of an outbreak, they may profit by buying options, rotating sectors, entering private equity positions, acquiring medical supply chains, or positioning sovereign funds. The economic harm is twofold: public biological vulnerability is converted into private gain, and the market becomes a battlefield of informational asymmetry.

However, financial gain is not equivalent to biological attack. The inference ladder is: abnormal gain $\Rightarrow$ suspicion $\Rightarrow$ foreknowledge $\Rightarrow$ actor-specific linkage $\Rightarrow$ convergence with biological/material evidence $\Rightarrow$ possible attribution.

This ladder prevents crisis profiteering from being confused with biological warfare.

20 International Relations and Geopolitics

The doctrine is compatible with realist and liberal-institutionalist theories of international relations. From a realist perspective, states must prepare for adversarial exploitation of biological uncertainty. From a liberal-institutionalist perspective, states need auditable rules, cross-border record preservation, and treaty-based penalties for obstruction. From a constructivist perspective, scientific consensus and origin narratives are themselves strategic objects that can be shaped through information warfare.

The doctrine therefore treats scientific evidence as a geopolitical asset. If early data vanish, the international system loses not only facts but deterrence. Evidence preservation is thus a form of defense.

21 False-Positive Controls

The model must avoid narrative overfitting. Therefore each empirical stream requires placebo tests:

1. placebo windows before W_F;
2. non-pandemic crisis windows;
3. ordinary earnings windows;
4. merger-rumor windows;
5. prior seasonal respiratory outbreaks;
6. routine procurement cycles;
7. index-rebalancing periods;
8. high-volatility market regimes.

The model should report p-values, false-discovery-rate adjusted q-values, Bayes factors, and posterior intervals.

Proposition 2 (Placebo discipline)**.** *If an anomaly appears equally in placebo windows and in W_F, then its attribution value is weak.*

Proof. Let E_W denote the anomaly in the focal window and E_P denote similar anomalies in placebo windows. If

$$\mathbb{P}(E_W \mid H_0) \approx \mathbb{P}(E_P \mid H_0)$$

and the focal anomaly is not distinguishable from placebo behavior, then the likelihood ratio of the anomaly under H_1 relative to H_0 is reduced. Hence the posterior update is weak. □

22 Conclusion

The Constitutional Doctrine of Biosecurity Evidence Preservation provides a disciplined response to strategic biological uncertainty. It does not presume biological warfare. It preserves the conditions under which biological warfare, lab accident, natural spillover, crisis profiteering, information warfare, or bureaucratic failure can be investigated.

The central mathematical rule is Bayesian and actor-specific:

$$\log \frac{\mathbb{P}(H_1(a) \mid E)}{\mathbb{P}(H_0 \mid E)} = \log \frac{\mathbb{P}(H_1(a))}{\mathbb{P}(H_0)} + \sum_{s \in \mathcal{S}} A_{s,a} \log LR_{s,a}.$$

The central constitutional rule is evidence preservation before attribution. The central evidentiary rule is that biological-warfare attribution requires at least two independent pre-D_0, costly-to-fake, actor-specific streams, including one biological/material stream. The central political rule is individualized remedy: punish named actors and obstructing institutions first; punish whole states only when command, protection, or knowing obstruction is shown beyond reasonable strategic doubt.

The doctrine protects the state from two symmetric errors: dismissing hostile biological action as impossible, and treating ambiguous crisis evidence as proof of war. It is therefore a constitutional, statistical, and strategic theory of how states should reason when biological truth itself becomes a battlefield.

References

[1] World Health Organization. WHO Timeline: COVID-19. 27 April 2020.

[2] World Health Organization. Pneumonia of unknown cause – China. Disease Outbreak News, 5 January 2020.

[3] Office of the Director of National Intelligence. Unclassified Summary of Assessment on COVID-19 Origins. 2021.

[4] Office of the Director of National Intelligence. Declassified Assessment on COVID-19 Origins. 2023.

[5] World Health Organization. WHO Scientific Advisory Group issues report on origins of COVID-19. 27 June 2025.

[6] World Health Organization. WHO issues its first emergency use validation for a COVID-19 vaccine. 31 December 2020.

[7] Lawrence O. Gostin. The Origins of Covid-19: Why It Matters. *New England Journal of Medicine*, 2023.

[8] Eugene F. Fama. Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, 25(2):383–417, 1970.

[9] A. Craig MacKinlay. Event studies in economics and finance. *Journal of Economic Literature*, 35(1):13–39, 1997.

[10] Donald B. Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701, 1974.

[11] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2009.

[12] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2006.

[13] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer, 2009.

[14] Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. CRC Press, 2013.

[15] Thomas C. Schelling. *Arms and Influence*. Yale University Press, 1966.

[16] Robert Jervis. *Perception and Misperception in International Politics*. Princeton University Press, 1976.

[17] Carl von Clausewitz. *On War.* Princeton University Press edition, 1976.

[18] Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.

[19] James M. Buchanan and Gordon Tullock. *The Calculus of Consent.* University of Michigan Press, 1962.

[20] Douglass C. North. *Institutions, Institutional Change and Economic Performance.* Cambridge University Press, 1990.

Glossary

Actor-network
A linked set of institutions, people, firms, labs, funds, state organs, contractors, and logistics channels used as the unit of inference.

Admissibility
The rule that evidence must be pre-D_0, costly to fake, and actor-specific before it can support attribution.

Biological/material stream
Evidence involving samples, serology, lab records, countermeasures, challenge studies, or pathogen-relevant biological materials.

Convergence rule
The rule that independent evidence streams must align before an actor-network is escalated from suspicion to attribution.

Counter-narration
A disciplined public or elite-facing message that demands transparency without premature accusation.

D_0 The first international disclosure date used as a public anchor.

Financial foreknowledge
Pre-disclosure financial positioning consistent with early knowledge of a biological shock.

Information-control evidence
Missing, suppressed, altered, restricted, or selectively disclosed early scientific data.

Strategic biological uncertainty
A state of uncertainty in which early biological evidence is missing or unreliable and the event has national-security implications.

State protection
Use of state power to block audits, threaten witnesses, manipulate courts, reward implicated networks, or prevent verification.

War-economy mobilization
State-firm coordination in vaccines, diagnostics, procurement, logistics, finance, and industrial production during a strategic crisis.

The End